Kalman Assimilation Model for Retrieving Time-Series Chlorophyll Content Based on Multisource Remote Sensing Images

Zhiyuan Wu, Lin Du, Dong Xing, Jian Yang, and Wei Gao

***Abstract*—The chlorophyll (Chl) is closely related to vegetation respiration and photosynthesis, and its long time-series data with high spatial resolution is significant for dynamics monitoring for vegetation growth status and environmental management. However, extant methodologies for long-term Chl content retrieval are encumbered with data lacunae and imbalance in high spatial resolution and continuous time-series coverage. To address these limitations, we have developed a novel Kalman filtering based Chl inversion assimilation mode (KFCAM) to synthesize high spatiotemporal resolution Chl content data in Wuhan. The KFCAM is used to: 1) obtain Chl time-varying rates from low-resolution (Landsat-8) images and high-resolution Chl information from Sentinel-2/SPOT5 images, simultaneously; 2) calculating the state error and Kalman gain; 3) this model integrates the high-resolution information into the time series by state updating model; and 4) synthesis continuous time-series images for Chl with 10-m resolution. The results demonstrate that: 1) the synthetic Chl image with KFCAM exhibits high accuracy with an R^2 of 0.906 $\pm$ 0.047 and a root-mean-square error (RMSE) of 1.801 $\pm$ 0.369; 2) the RMSE values reveal a 12% and 33% reduction compared to the spatiotemporal gap filling (STGF) and temporal adaptive reflectance fusion model spatial and temporal adaptive reflectance fusion model (STARFM), respectively; and 3) furthermore, in comparative experiments that considered different assimilation time intervals and changes in land cover, KFCAM outperformed two reference models in terms of stability and anti-interference capability. It provides a novel means for large-scale and global Chl detection.**

***Index Terms*—Chlorophyll (Chl), data assimilation, Kalman filter, time series.**

I. Introduction

THE long time-series chlorophyll (Chl) content with high spatial resolution is closely related to the process of respiration and photosynthesis of vegetation, which can indicate the dynamical changes in vegetation growth status by synthesizing atmospheric carbon dioxide and water taken up by the vegetation roots [1]. Furthermore, the Chl is one of the important biochemical pigments that can continually convert light energy into chemical energy stored as carbohydrates together with other pigments. Therefore, the rapid and accurate monitoring of vegetation Chl content across continuous spatial–temporal scales is of great significance for environmental and even ecological research [2]. The utilization of high spatial–temporal resolution remote sensing imagery enables large-scale, nondestructive monitoring of Chl content, while it is affected by special sensor errors and environmental factors [3]. Particularly, the incomplete coverage of images over time remains a challenge in temporal series Chl content retrieval [4].

Previous research on Chl content retrieval is based on linking its relationship with the reflectance from multiband images [5], and different spectral indices derived from reflectance band commonly have varying sensitivity to actual Chl concentration [6]. Sun et al. [7] constructed an index using the near-infrared (NIR), red, and green bands from MERIS data to inversely retrieve Chl images at seven-day intervals. However, the spatial resolution was limited to 30 m, and the time series spanned only six months. With extended temporal scales, Sherman et al. [8] constructed a weighted index with five more spectral bands (443, 490, 560, 674, and 681 nm) from Sentinel-3 data, achieving the retrieval of continuous monthly Chl images from January 2016 to December 2021. Except for the decreased temporal resolution, the spatial resolution was enhanced to 10 m, and the time series was extended to cover five years. These studies are basically with single-source remote sensing data and have been limited by satellite revisit cycles and atmospheric conditions [9], leading to data gaps in the required time series. Many studies have proposed algorithms to fill in long-term Chl time-series images to solve the problem of missing data, including machine learning and spatiotemporal information interpolation. However, these methods have certain difficulties in accurately inverting the true spatiotemporal changes of Chl [10], [11], [12]. Currently, the most effective existing algorithm for interpolating and supplementing Chl time-series data is the spatiotemporal gap filling (STGF), which has a strong performance in maintaining time continuity. Zhou et al. [13] found that the correlation between the continuous Chl time-series data synthesized from eight days of MODIS composite Chl products and the measured values was as high as 0.97. Nevertheless, it is important to note that these methods may be limited by various conditions, being hard to balance the high temporal–spatial resolution [14].

For this purpose, multiobservation data fusion from different satellites have become crucial in satellite imagery analysis, particularly for addressing data gaps [15]. For

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instance, Xiao et al. [16] employed Sentinel-2 imagery to compensate for the missing 700–710-nm band in Landsat-8/9, successfully synthesizing 10-m resolution Chl images; nonetheless, they struggled to achieve ideal results over extended temporal scales. Meanwhile, Wang and Chen [17] collected data from four satellite sensors (MERIS, OLCI, OLI, and MSI) to enhance the Chl image time series while maintaining a 10-m spatial resolution. Unfortunately, systematic errors among these different sensor data resulted in a lower correlation between the synthesized results and observed data ($R^2 \leq 0.8$). Moreover, these errors arise from variations in sensor parameters and the spectral coverage of different bands, pixel size, and the mixed pixel effect [18], [19], [20]. To mitigate these errors, Gao et al. [21] developed a spatial and temporal adaptive reflectance fusion model (STARFM) that multiple pairs of MODIS–Landsat observations and multiple MODIS images with predicted times were used to generate a 30-m resolution surface reflectance image (prediction error as low as 0.01). This approach was subsequently extended to other sensor data. To date, STARFM has become the most widely used and mature method for fusing fine spatial resolution with high-frequency temporal information. However, the correlation between images is heavily influenced by atmospheric conditions, and its accuracy immediately decreases when the correlation weakens. Ma et al. [22] performed linear spectral unmixing on coarse-resolution MODIS pixels and integrated the derived reflectance change rates for different land-cover types into Landsat imagery to synthesize a long-term time series of Landsat images. While this approach alleviates the need for high-quality observational data to some extent, it heavily relies on the unmixing accuracy of mixed pixels and performs poorly for pixels with significant land-cover changes [23]. In summary, current fusion models do not focus on quantitatively evaluating how uncertainty and stability in time-series predictions change over time, which is also a necessary step in the quantitative analysis of Chl time series.

Many researchers have proposed data assimilation methods to reduce the aforementioned errors and improve model prediction accuracy, rather than relying solely on information fusion [24]. For example, Tang and Huang [20] integrated simulated annealing (SA) and genetic algorithms (GAs) into support vector machines (SVMs) to establish an SVM-based collaborative model (Col-SVM). Then, they simulated Chl concentrations in Donghu Lake at different time periods using various datasets. This method merely integrates centralized algorithms through machine learning without exploring the underlying mechanisms of the model's operation. Mattern et al. [25] coupled biogeochemical numerical models with physical ocean circulation models to assimilate the NPZD model and the NEMURO model with a 3-D physical ocean circulation model based on the regional ocean model system (ROMS). They combined Chl satellite observations with physical observations using 4-D-Var state estimation to improve the accuracy of Chl data. However, the estimation of 4-D-var requires the prior specification of an accurate and reliable background error value; otherwise, it may lead to the implementation of unrealistic corrections to unobserved variables. To circumvent the potential influence of model state variables and Gaussian background errors, Guo et al. [26] have proposed a data assimilation scheme that utilizes particle filtering to assimilate Chl concentrations. This method obtains a substantial enhancement in the precision of forecasts regarding Chl concentrations and can be seamlessly integrated into a wide range of nonlinear dynamic systems. Nevertheless, it is important to note that particle filter-based assimilation systems are incapable of updating system states in real time. In instances where the intermediate simulation results of the model exhibit significant deviation from observations, modifications and adjustments to the model parameters must be executed subsequent to the completion of the simulation. This is detrimental to the time-series prediction of Chl. Research has demonstrated that Kalman filtering (KF) can perform real-time updates of state estimates and dynamically estimate observation errors during model executions. Furthermore, the KF is a widely used algorithm for sensor fusion and data integration [27], which can effectively overcome some errors generated by different data sources [28]. Thus, it has been utilized pervasively in domains of remote sensing image filtering, data assimilation, and time-series analysis [29], [30]. Consequently, KF-based assimilation methods have emerged as a promising new direction for improving Chl time-series prediction. Zhou and Zhong [31] developed a Kalman filter reflectance fusion model (KFRFM), which was the first to employ the Kalman algorithm to integrate MODIS time series into Landsat data, synthesizing a long-term time series of Landsat reflectance data. Nevertheless, these studies have not directly applied the KF algorithm to predict Chl time-series data, and retrieving Chl content from surface reflectance is subject to substantial uncertainties [32]. However, by constructing canopy models based on scattering coefficients or feature-extraction methods [33], [34], scientists can achieve precise, controllable inversion of surface reflectance to Chl [35]. Therefore, developing coupled inversion and KF-based assimilation models to predict Chl spatiotemporally is an efficient and feasible solution.

Inspired by the challenges in Chl retrieval, we propose a KF-based data assimilation model. The concept of Kalman data assimilation for spatiotemporal data can be directly applied to surface Chl content. However, this requires establishing an accurate mapping between reflectance and actual Chl content before data assimilation can be implemented, which directly enhances the model performance in various environmental conditions [36]. Compared to empirical methods based on vegetation index inversion, establishing a lookup table (LUT) allows for the retrieval of the most similar records from the LUT for each new set of observational data [37], thereby creating an accurate mapping relationship between Chl and bands [38]. For instance, Wang et al. [39] used Landsat-8 data to establish an LUT based on the PROSAIL model, successfully inverting a detailed leaf area index (LAI), although the spatial resolution was relatively low at 30 m. In contrast, Qian and Liu [40] developed an LUT for the PROSAIL model and employed Sentinel-3 imagery to invert Chl images with a higher spatial resolution of 10 m. The Chl concentrations obtained from this method align more closely with real conditions compared to

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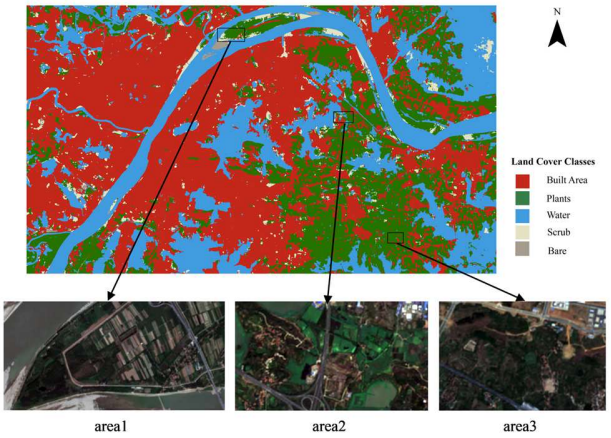


Fig. 1. Study area and its land-cover map.

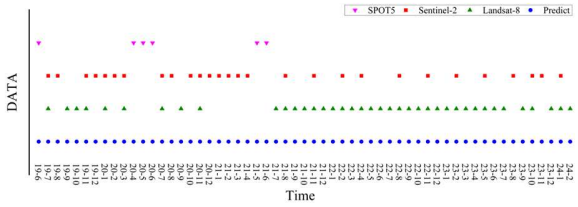


Fig. 2. Distribution of remote sensing imagery in time series.

TABLE I
LAND-COVER TYPES AND THEIR PERCENTAGE WITHIN THE STUDY AREA

	Land cover class	%
Primarily vegetated areas	Plants	22.55
	Scrub	2.79
Primarily non-vegetated areas	Built Area	47.13
	Water	27.03
	Bare	0.5

those derived from empirical models. Furthermore, Sun et al. [41] applied the LUT-based radiative transfer model (RTM) to the HSL-LiDAR system, expanding the data and application scenarios of the LUT. This study innovatively integrates the Kalman algorithm with the LUT inversion method based on RTMs, proposing a coupled model for high spatiotemporal inversion of Chl content through the fusion of multisource spatiotemporal remote sensing data. By contrast, Kalman filtering based Chl inversion assimilation mode (KFCAM): 1) defines the model state as Chl concentration (not reflectance); 2) obtains high-resolution observations via PROSAIL-LUT inversion; and 3) explicitly model observation and process uncertainty at the pixel level—including dynamic adjustment of observation covariance used in the Kalman gain. These three changes together produce the improved stability and reduced errors.

In this study, a PROSAIL RTM is utilized to create an LUT, which converts multisource remote sensing images (Landsat-8, Sentinel-2, and SPOT5) into Chl maps. Landsat-8 provides coarse resolution data, while Sentinel-2 and SPOT5 offer fine resolution imagery. The KF extracts the rate of change from the coarse-resolution Chl image, which is then used to predict the fine-resolution Chl (Fine-Chl) image. This rate is added to the predicted image to generate the final Chl map. Additionally, we assessed the impact of land-cover changes and observation intervals on the results. This article is structured as follows. Section II introduces the data and methodology, Section III presents the results, Section IV analyzes the results, and Section V concludes this study.

II. METHODOLOGY

A. Study Area and Imagery Data

The research area is concentrated in the Yangtze River and Donghu areas within Wuhan, Hubei Province, China (Fig. 1). Wuhan is classified as a subtropical monsoon climate, characterized by high precipitation levels, particularly from June to September [42], [43]. The watershed area is extensive, encompassing numerous lakes and water bodies, constituting 27.03% of the study area. The urbanization rate in this region is as high as 80.29% [44], resulting in large areas of built-up zones with no vegetation cover in remote sensing imagery, leading to gaps in the Chl images. A rich variety of vegetation types is the characteristic of parks and scenic areas (Table I), and the growing season for vegetation in the study area typically extends from March to August each year [45]. The selection of three subareas for subsequent analysis and discussion was based on the presence of distinct vegetation types and varying levels of vegetative cover, including grassland (area1), aquatic plants (area2), and trees (area3).

Three types of satellite imagery with different resolutions are utilized in this study. Landsat-8 OLI imagery is employed to generate coarse-resolution Chl images with a spatial resolution of 30 m. The primary benefit of the Landsat-8 satellite is its ability to provide continuous imaging of the study area over an extended period, with a maximum time coverage of up to 16 days. A total of 41 images are selected, with one image per month from January 2019 to February 2024. Sentinel-2 MSI images are utilized to generate high-resolution Chl images with a resolution of 10 m. These images are also employed as supplementary observation images to fill gaps in the Chl time series when Landsat-8 images are unavailable. The analysis is conducted on a total of 29 images for Sentinel-2 during the time range. Furthermore, a selection of six 10-m resolution SPOT5 images is utilized as a substitute for Sentinel-2 data during periods of unavailability, thereby facilitating the production of high-resolution Chl images. It is noteworthy that all the selected remote sensing images exhibit cloud cover of less than 10% (Fig. 2). Before applying the KFCAM, we need to preprocess all remote sensing images (e.g., atmospheric correction) and perform calibration to ensure that the geographic coordinate systems of all remote sensing images are consistent.

B. Chl Inversion With PROSAIL-Based LUT

Unlike earlier KF-based fusion methods that operate in the surface-reflectance domain, KFCAM performs assimilation in the Chl state space by combining PROSAIL-based LUT

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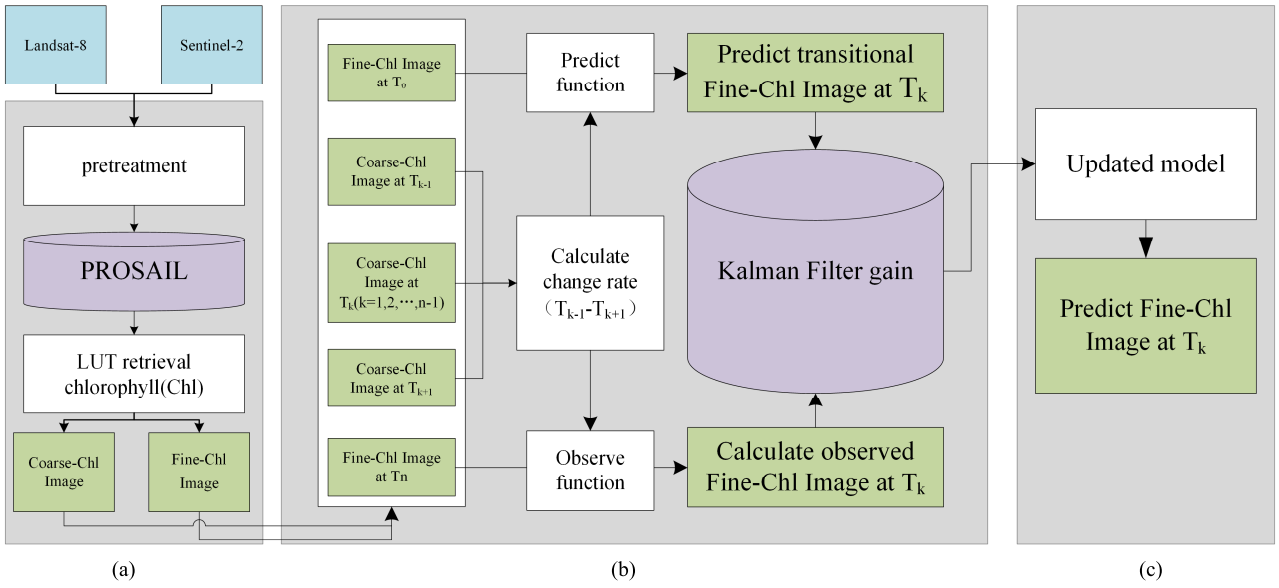


Fig. 3. Main technical workflow diagram of KFCAM. (a) Satellite image inversion Chl image with LUT. (b) Kalman state prediction and observation model. (c) State updating.

TABLE II
STATISTICAL CHARACTERISTICS OF THE PROSAIL MODEL INPUTS USED TO GENERATE LUTS

Parameters	Unit	Min	Max	Parameter step	Mean	Std
N	-	1.0	3.5	0.1	1.6	0.3
C_{ab}	μg/cm^2	0	100	1	34.59	18.87
C_{ar}	μg/cm^2	0	30	1	8.51	3.92
C_{anth}	μg/cm^2	0	40	1	8.8	7.1
C_w	μg/cm^2	0.00005	0.045	0.00005	0.0129	0.0073
C_m	μg/cm^2	0.002	0.04	0.002	0.0077	0.0035
LAI	-	0	7	0.1	3.5	2.5
ALA	°	30	80	5	60	20
hspot	-	0	1	0.1	0.45	0.6
psoil	-	0	1	0.1	0.5	0.5

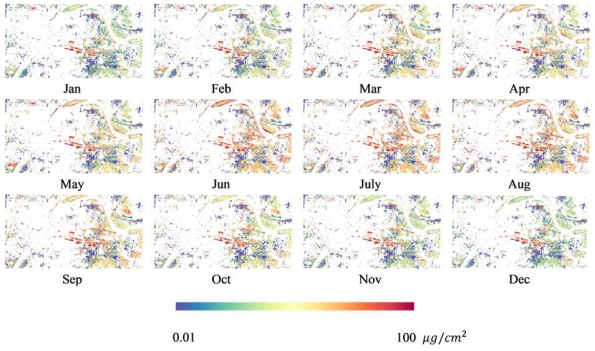


Fig. 4. Synthetic Chl images for 2022.

inversion with KF; this enables direct use of RTM-constrained Chl observations and dynamic, observation-level uncertainty modeling.

The inputs of PROSAIL model encompass leaf parameters [leaf structure parameter (N), Chl content (C_{ab}), carotenoid content (C_{ar}), anthocyanin content (C_{anth}), equivalent water thickness (C_w), and leaf mass per area (C_m)], canopy parameters [LAI, average leaf angle (ALA), and hotspot parameter (hspot)], and soil reflectance (psoil) [46] (Table II). The size of the LUT was set to 10 000, and the reasonable combinations of parameter were generated after the sensitivity analysis within their corresponding ranges based on the public database [47].

For robust cross-sensor matching, PROSAIL-simulated spectra were convolved with each sensor's published spectral response functions (SRFs) to produce sensor-specific LUT entries. During inversion, measured multispectral reflectance is compared to the SRF-convolved LUT entries, ensuring that bandpass differences are accounted for in the observation generation step.

The cost function serves as a minimization criterion to quantify the consistency between simulated spectra and observed values, significantly reducing redundant results and errors in the LUT [48]. Therefore, selecting the optimal cost function is crucial for effectively handling data with varying distribution characteristics and types of error distribution. In this study, we consider four of the most widely used and best-performing cost

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TABLE III
R^2 FOR THE COMPARISON OF THE INVERSE CHL OF MULTIPLE OPTIMAL SOLUTIONS WITH THE MEASURED DATABASE CHL

Number of solutions	LSE	GM	LP	MC
1	0.462	**0.464**	0.353	0.441
10	**0.532**	0.523	0.403	0.451
50	**0.568**	0.542	0.421	0.440
100	0.548	**0.573**	0.449	0.427
200	**0.559**	0.542	0.458	0.428
500	**0.561**	0.558	0.476	0.449
1000	**0.570**	0.566	0.480	0.485

functions for comparison [see (1)–(4)], where R_{mea} and R_{LUT} are the measured and simulated reflectances at wavelength λ_i, respectively, and n is the number of wavelengths.

The least-squares estimation [LSE, (1)] has been shown well performance when addressing Gaussian distribution errors in the cost function [49]. Two other least-squares-based estimators, the Geman–McClure [GM, (2)] and the Laplace distribution [LP, (3)], perform exceptionally well when dealing with symmetric errors [50]. Differently, the minimum contrast distance [MC, (4)] aims to minimize the distance between parametric and nonparametric spectral densities [51]

$$\text{LSE} = \sum_{\text{i=1}}^{n} \left(R_{\text{mea}}(\lambda_i) - R_{\text{LUT}}(\lambda_i)\right)^2 \tag{1}$$

$$\text{GM} = \sum_{\text{i=1}}^{n} \frac{\left(R_{\text{mea}}(\lambda_i) - R_{\text{LUT}}(\lambda_i)\right)^2}{1 + \left(R_{\text{mea}}(\lambda_i) - R_{\text{LUT}}(\lambda_i)\right)^2} \tag{2}$$

$$\text{LP} = \sum_{1}^{n} \left|R_{\text{mea}}(\lambda_i) - R_{\text{LUT}}(\lambda_i)\right| \tag{3}$$

$$\text{MC} = \sum_{i=1}^{n} \left(\log\left(\frac{R_{\text{LUT}}(\lambda_i)}{R_{\text{mea}}(\lambda_i)} + \frac{R_{\text{mea}}(\lambda_i)}{R_{\text{LUT}}(\lambda_i)}\right)\right). \tag{4}$$

We conducted optimal matching between 10 000 PROSAIL-simulated spectra and measured spectra from our database by applying several different loss functions. For each loss function, we identified the best-matching database spectrum for every simulated spectrum. We, then, ranked these matches by loss value and, for each loss function, extracted the Chl content of the best 1, 10, 50, 100, 200, 500, and 1000 simulated spectra. Finally, we compared these extracted Chl values to those of their corresponding matched spectra in the database to assess retrieval accuracy (Table III). GM gives the best results when only optimal solutions are considered, but R^2 of LSE reaches optimality when there are more than 200 solutions. Therefore, we chose to employ LSE to construct the LUT for the inversion of Chl content.

C. Kalman Filter Algorithm

KF integrates observations, models, and their respective uncertainties to estimate the state of the process by minimizing the average squared error [52]. The process typically consists of two steps. The first step, "prediction," involves the Kalman filter generating estimations of the current state variables along with their associated uncertainties. The second step, "updating," refines these estimations using a weighted average that incorporates new observed measurements, assigning greater weight to estimations with higher certainty

$$X_t = \Phi_{t,t-1} X_{t-1} + \Gamma_{t,t-1} U_{t-1} + W_{\text{t}} \tag{5}$$

$$Z_t = H_t X_t + V_t. \tag{6}$$

KF is typically expressed as a continuous process over time, where the variable state of the model at time t (X_t) can be evolved from the state at time $t-1$(X_{t-1}). This evolution is processed with the state transition matrix $\hat{O}_{t,t-1}$ [see (5)]. Γ is the control input matrix used to control U_{t-1}, and W_t is the evolution process noise, which follows a normal distribution with covariance Q_t: $W_t \sim N(0, Q_t)$; through mapping state space to the observation space by observation model H_t, the observation at time t (Z_t) is expressed as (6), where V_t is the observation noise with covariance S_t that follows a normal distribution N: $V_t \sim N(0, S_t)$. When predicting the state, there is a state estimation covariance

$$P_{t,t-1} = \Phi_{t,t-1} P_{t,t-1} \Phi_{t,t-1}^{\text{T}} + Q_t. \tag{7}$$

Both the state estimate $\hat{X}_t$ and the posterior state estimation covariance (P_t) are updated, in which K_t represents the optimal Kalman gain

$$\hat{X}_t = \hat{X}_{t,t-1} + K_t\left[Z_t - H_t \hat{X}_{t,t-1}\right] \tag{8}$$

$$P_t = P_{t,t-1} - K_t R_t K_t^{\text{T}} \tag{9}$$

$$K_t = P_{t,t-1} H_t^{\text{T}} R_t^{-1} \tag{10}$$

$$R_t = S_t + H_t P_{t,t-1} H_t^{\text{T}}. \tag{11}$$

To validate the KF, we need two pairs of images, such as those from Landsat-8 and Sentinel-2 at times T_0 and T_n, as well as the predicted Landsat-8 image (Coarse-Chl) at the prediction time $T_k (k = 1, 2, \ldots, n-1)$. The Fine-Chl image at time T_0 is used for state prediction, while the Fine-Chl image at time T_k is used to update the state through the observation model, then implement the KFCAM to predict the Fine-Chl content data for time T_k. The major process steps are illustrated in Fig. 3.

Compared with the standard Kalman formulation and existing Kalman-based spatiotemporal fusion algorithms, KFCAM treats high-resolution images as direct observations and incorporates the actual Chl change rate together with a state-transition matrix to jointly govern state prediction and update. We introduce novel uncertainty matrices, error-management functions, and update operators that distinguish KFCAM from

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prior Kalman-based fusion methods and enable quantitative evaluation of the model's operational uncertainty. The observation covariance R_t is estimated from the LUT matching residuals (the difference between measured reflectance and SRF-convolved PROSAIL best matches, translated into Chl-space uncertainty via the inversion Jacobian), and process noise Q_k is derived from coarse-scale change-rate variance aggregated over neighboring pixels/clusters.

1) State Prediction: The covariance of the Fine-Chl image at the starting time T_0 serves as an uncertainty estimate for predicting the Fine-Chl image at time T_k. This prediction is based on the difference between the coarse-resolution Chl images at times T_{k-1} and T_k, along with the change rate at time T_k.

A challenge in the state prediction step is that the pixels in the Fine-Chl and Coarse-Chl images do not correspond directly. Additionally, during preprocessing, extracting the vegetation area may result in regions with missing values in the image. These issues contribute to errors in state prediction. To address this, the pixels in each Coarse-Chl image are grouped into h clusters, with each cluster containing all the Fine-Chl pixels corresponding to a single Coarse-Chl pixel. Continuous pixels are grouped into the same cluster.

The Chl change from T_{k-1} to T_k can be expressed as follows:

$$\hat{C}_{k,k-1} = C_{k-1} + (T_k - T_{k-1})\, r_{k-1}. \tag{12}$$

The Chl content $\hat{C}_{k,k-1}$ at T_k can be viewed as the real Chl at T_{k-1} (C_{k-1}) changing from T_{k-1} to T_k with a change rate at T_{k-1} (r_{k-1}). The uncertainty matrix of Chl is represented by covariance. At T_k, the uncertainty $\hat{P}_c^{k,k-1}$ is updated in real time from P_c^{k-1} at T_{k-1} using the state-transition matrix of Chl content $F_{k,k-1}$, ensuring dynamic and accurate error propagation

$$\hat{P}_c^{k,k-1} = F_{k,k-1} P_c^{k-1} F_{k,k-1}^{\mathrm{T}} + Q_k \tag{13}$$

$$F_{k,k-1} = \frac{\partial C_k}{\partial C_{k-1}} Q = \sigma_L^2 + \sigma_B^2. \tag{14}$$

Assuming that C_B^k and C_B^{k-1} are Coarse-Chl datasets on two different times T_k and T_{k-1}, and the uncertainty in variance of Fine-Chl and Coarse-Chl observation are σ_L^2 and σ_B^2, we can calculate the rate of change r_B and uncertainty $\sigma_{r_B}^2$ for each pixel in the coarse-resolution Chl image as follows:

$$r_B = \frac{\left(C_B^k - C_B^{k-1}\right)}{T_k - T_{k-1}} \tag{15}$$

$$\sigma_{r_B}^2 = \frac{2\sigma_B^2}{\Delta t^2}. \tag{16}$$

In terms of data organization and pixel alignment, we referred to the formula of KFRFM [31]. For each pixel in the coarse-resolution Chl image, the increase in Chl content is the weighted sum of the Chl content of each Fine-Chl pixel within the coarse pixel

$$\Delta C_B(x_i, y_i) = \sum_{i=1}^{h} f(x_i, y_i) \cdot \Delta C_L \tag{17}$$

$$f(x_i, y_i) = \frac{N(x_i, y_i)}{N} \tag{18}$$

where $f(x_i, y_i)$ is the proportion of fine-resolution pixels within the coarse-resolution pixel at this coordinate. This is determined by the ratio of the number of fine-resolution pixels within the cluster $N(x_i,\ y_i)$ to the total number of fine-resolution pixels N within the coarse-resolution pixel. According to (17), we can derive the functional relationship between r_B and $f(x_i, y_i)$

$$\begin{bmatrix} r_B(x_1, y_1) \\ \vdots \\ r_B(x_i, y_i) \\ \vdots \\ r_B(x_n, y_n) \end{bmatrix} = \begin{bmatrix} f_1(x_1, y_1) & \cdots & f_i(x_1, y_1) & \cdots & f_h(x_1, y_1) \\ \vdots & & \vdots & & \vdots \\ f_1(x_i, y_i) & \cdots & f_i(x_i, y_i) & \cdots & f_h(x_i, y_i) \\ \vdots & & \vdots & & \vdots \\ f_1(x_n, y_n) & \cdots & f_i(x_n, y_n) & \cdots & f_h(x_n, y_n) \end{bmatrix} \begin{bmatrix} r_1 \\ \vdots \\ r_i \\ \vdots \\ r_h \end{bmatrix} \tag{19}$$

$$\hat{r} = \left(A^{\mathrm{T}} W_B A\right)^{-1} A^{\mathrm{T}} W_B R_B \tag{20}$$

$$P_r = \hat{\sigma}_r^2 \left(A^{\mathrm{T}} W_B A\right)^{-1}. \tag{21}$$

We can obtain the Fine-Chl change rate $\hat{r}$ and its covariance matrix estimate P_r [53] where

$$A = \begin{bmatrix} f_1(x_1, y_1) \cdots f_i(x_1, y_1) \cdots f_h(x_1, y_1) \\ \vdots \qquad\qquad \vdots \qquad\qquad \vdots \\ f_1(x_i, y_i) \cdots f_i(x_i, y_i) \cdots f_h(x_i, y_i) \\ \vdots \qquad\qquad \vdots \qquad\qquad \vdots \\ f_1(x_n, y_n) \cdots f_i(x_n, y_n) \cdots f_h(x_n, y_n) \end{bmatrix} \tag{22}$$

$$R_B = \begin{bmatrix} r_B(x_1, y_1) \\ \vdots \\ r_B(x_i, y_i) \\ \vdots \\ r_B(x_n, y_n) \end{bmatrix},\ V_B = A\hat{r} - R_B \tag{23}$$

$$\hat{\sigma}_r^2 = \frac{V_B^{\mathrm{T}} W_B V_B}{n - \mathrm{h}},\ W_B = Q_B^{-1},\ Q_B = \sigma_{r_B}^2 I. \tag{24}$$

The variance matrix Q_B of the measurement R_B is calculated from the uncertainty of the coarse-resolution Chl content σ_B^2. $\hat{\sigma}_r^2$ is the unit variance estimation from the residuals of the observation, and I is a unit matrix.

When the updated synthetic Fine-Chl at T_{k-1} is used as the first part of the reflectance transition model, P_c is the uncertainty in the variance of the updated synthetic Fine-Chl image.

2) Observation Model: Based on the basic principles of KF in (5) and (6), the KFCAM generates the system state transition model using a pair of Fine-Chl and Coarse-Chl data at T_0. The observation model uses Fine-Chl retrieved from actual high-resolution images at T_n as the observation. Compared with observations synthesized from change-rate calculations and state estimates, these direct high-resolution

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observations better constrain observation error and thereby improve assimilation accuracy.

The Chl observation $\hat{C}_{k,n}$ at T_k is obtained by updating the Fine-Chl C_n at T_n using the change rate r_n from T_k to T_n. The corresponding observation covariance $\hat{P}_c^{k,n}$ is propagated via the state-transition matrix $F_{k,n}$, so that observation error is dynamically regulated by the state evolution

$$\hat{C}_{k,n} = C_n + (T_k - T_n)\, r_n \tag{25}$$

$$\hat{P}_c^{k,n} = F_{k,n} P_c^n F_{k,n}^{\mathrm{T}} + Q_k. \tag{26}$$

Similarly, P_c^n is the uncertainty matrix of the Fine-Chl at T_n.

3) State Updating: $R_{k,n}$ serves as the observation model for backward updating, and $R_{k,k-1}$ serves as the state model for forward prediction. The optimal Kalman gain K_k is calculated based on the uncertainties of the two matrix models. Since $H_k = I$ in our case, the system state transition model is, then, updated accordingly as follows:

$$R_k = \hat{P}_c^{k,n} + \hat{P}_c^{k,k-1} \tag{27}$$

$$K_k = \hat{P}_c^{k,k-1} R_k^{-1}. \tag{28}$$

Note that because the observation uses Fine-Chl retrieved from actual high-resolution imagery and the observation error is dynamically adjusted according to state evolution, a quantitative observation uncertainty matrix $\hat{P}_c^{k,n}$ can be incorporated into the state-update step (the following equation). The actual formula can be obtained as follows:

$$C_k = \hat{C}_{k,k-1} + K_k \left[\hat{C}_{k,n} - \hat{P}_c^{k,n} - \hat{C}_{k,k-1}\right]. \tag{29}$$

After propagation and algebraic simplification, the predictive uncertainty matrix of the synthesized Fine-Chl image (excluding observation noise) is obtained and denoted as P_k

$$P_k = (I - K_k) R_k - K_k R_k K_k^{\mathrm{T}} = \hat{P}_c^{k,k-1} \left[I - K_k^{\mathrm{T}} H_k\right]. \tag{30}$$

This uncertainty supports the construction of a quantitative metric, model coverage (MCOV), for assessing the model's overall operational uncertainty.

D. Experimental Design

The comprehensive performance of the proposed KFCAM in temporal Chl inversion was evaluated against two comparative models: STARFM and STGF (Table IV). The former is a mature spatiotemporal-fusion model widely used in multisensor data fusion that focuses on restoring spatial details in high-resolution imagery; the latter is a recently proposed mainstream approach for time-series completion, emphasizing interpolation of missing observations and the preservation of long-term temporal consistency. Comparing these two models enables a comprehensive assessment of KFCAM's advantages in both spatial-detail recovery and temporal continuity.

Additionally, we conducted comparative experiments using different assimilation time intervals and examined the impact of land-cover changes. This investigation assessed how varying time intervals affect the prediction accuracy of high-resolution images and how land-cover changes influence the predicted Chl images over time. Furthermore, we tested the stability of the model under these conditions.

For validation, we preferentially used cloud-free Sentinel-2 scenes as an independent reference. Where Sentinel-2 was unavailable, SPOT5 inversions were used as an alternate validation source. To test for bias arising from this alternation, we performed cross-validation experiments in which Sentinel-2 and SPOT5 were alternately used as the validation target. Root-mean-square error (RMSE) differences between these validations were negligible (0.004–0.104, Table V), supporting the conclusion that SPOT5-based validation does not introduce material bias in our experiments.

The prediction experiments consist of two main components: time-series prediction and spatial prediction. The first part involves further predicting synthetic Chl images at 15-day intervals following the initial time-series prediction. Due to the limitations of existing satellite remote sensing images in terms of national coverage, the second part of the experiment focuses on the nationwide prediction of Chl images. The high-resolution Chl images synthesized by the KFCAM model offer a comprehensive view of Chl content at the national scale, with a resolution of 10 m.

III. Results

The experiment produced a time series of 57 Chl images—one per month from June 2019 to February 2024—for the full study area. These images were then cropped to derive Chl maps for three smaller subregions. To examine seasonal dynamics, the Chl assimilation outputs for 2022 were analyzed in detail. Finally, in September 2022, we carried out a comparative evaluation of three synthesis methods (KFCAM, STGF, and STARFM) against Sentinel-2 clear-sky observations, employing both qualitative visual assessment and quantitative scatterplot analysis. By implementing the predictive experiment, we enhanced temporal resolution—reducing the interval to 15 days—to generate 113 synthesized Chl images, including comprehensive national-scale Chl composites.

We emphasize that the validation results shown use Sentinel-2 clear-sky scenes; repeating the very similar comparisons with SPOT5 as the validation reference yields essentially identical RMSE (Table V), indicating that the use of SPOT5 for validation does not bias the reported performance metrics.

A. Assimilated Chl Content in Some Urban Areas of Wuhan in the Year of 2022

Through KFCAM assimilation, we observe that the high values of vegetation Chl are concentrated around the Donghu Scenic Area in the central part of the study area, while the low values are located in the southern and northwestern parts of the study area. Furthermore, the Chl content demonstrates a regular seasonal variation trend (Fig. 4), increasing from February to July and subsequently decreasing from August to December. The Chl content typically peaks in July or August and reaches its lowest point around January.

During the execution of the model, the 30-m time-series Chl data provide the temporal rate of change, while the spatial accuracy is predominantly influenced by the inserted 10-m Chl imagery. This finding is corroborated by Fig. 5,

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TABLE IV
RMSE FROM KFCAM, STGF, AND STARFM

	Time												
	22-8	22-9	22-10	22-11	22-12	23-1	23-2	23-3	23-4	23-5	23-6	23-7	23-8
KFCAM	1.46	1.45	1.58	1.68	1.47	1.24	1.76	1.53	1.68	1.46	1.66	1.47	1.31
STGF	1.91	1.87	1.68	2.08	1.57	1.77	2.34	2.12	2.25	2.85	1.97	2.37	2.15
STARFM	2.11	2.13	1.85	2.19	1.78	2.05	2.47	2.31	2.49	2.95	2.26	2.62	2.36

TABLE V
COMPARISON OF RMSE WITH DIFFERENT VALIDATION DATA

Validation data	Time												
	22-8	22-9	22-10	22-11	22-12	23-1	23-2	23-3	23-4	23-5	23-6	23-7	23-8
Sentinel-2	1.463	1.459	1.623	1.686	1.481	1.247	1.751	1.551	1.684	1.458	1.664	1.462	1.323
SPOT5	1.467	1.452	1.604	1.674	1.470	1.351	1.772	1.541	1.677	1.469	1.725	1.511	1.316

which demonstrates that the synthesized Chl results from all three models align more closely with the 10-m observational data.

KFCAM demonstrates the best performance, exhibiting the greatest similarity to the observed image in terms of spatial distribution details. The STARFM model produces areas of high Chl values that are lower than those observed, particularly in area1 [Fig. 5(a)] and near the water body [Fig. 5(b)], where abnormally low values are noted. The STGF model outperforms STARFM in most areas but exhibits significantly low performance in the grass type [Fig. 5(a)] and near the built-up land [Fig. 5(c)] compared to the observed impacts. Moreover, the results in areas exhibiting low Chl values are also lower than those observed images.

B. Comparison of Spatial Distribution

The KFCAM model demonstrated the highest spatial distribution accuracy of the three models [attaining $R^2 = 0.922$ and RMSE = 1.43 in area3, Fig. 6(c)], while STARFM exhibited the lowest performance in area1 [$R^2 = 0.776$ and RMSE = 4.47, Fig. 6(a)]. The overall distribution trend indicates that the distribution of high values of Chl for STGF in area2 and area3 deviates from the 1:1 line and approaches the observed values, suggesting that the observed values exceed those of STGF in the high value region, which is consistent with the previous results. The distribution of the scatters of STARFM deviates toward the observed values in all the study areas, suggesting that the results of the Chl of the inversion of STARFM are more similar to those of the observations compared to the Chl that is skewed lower. Furthermore, the KFCAM results demonstrated two areas of Chl concentration, 10–20 and 50–60 μg/cm^2, while STGF and STARFM exhibited only one area of Chl concentration, which was lower than the observed value in the range of 0–10 μg/cm^2. These findings are consistent with the conclusions drawn from previous visual interpretation. Overall, the KFCAM model exhibited greater predictive performance, with RMSE reductions of 10%–20% compared to STGF and 20%–40% compared to STARFM [Fig. 6(b) and (c)].

C. Model Uncertainty

There are many quality metrics for evaluating data quality of the blended images against observations, where we selected three commonly used quality metrics for assessment and model comparison: the Pearson correlation coefficient (PCC), quality index (QI), RMSE, and coefficient of determination (R^2). The PCC and QI reflect the degree of correlation and similarity of the structures between the predicted and reference images, respectively, with values closer to one signifying a strong linear relationship and structural similarity [54], [55]. A higher R^2 and smaller RMSE indicate a closer match between the predicted and reference images [56].

$$\mathrm{PCC} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} \tag{31}$$

$$\mathrm{QI} = \frac{4\sigma_{xy}\bar{x}\bar{y}}{\left(\sigma_x^2 + \sigma_y^2\right)\left(\bar{x}^2 + \bar{y}^2\right)} \tag{32}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2} \tag{33}$$

$$R^2 = 1 - \frac{\sum (y_i - x_i)^2}{\sum (y_i - \bar{y})^2} \tag{34}$$

where $\bar{x}$ and $\bar{y}$ are the mean values of a synthetic Chl image (x) and the corresponding observed Chl image (y); σ_x^2, σ_y^2, and σ_{xy} are their variances and covariance. There are a total of n pixels for each image.

A quantitative evaluation of the models' performance during August 2022–August 2023 reveals that the KFCAM model demonstrates the optimal performance, followed by the STGF model, with the STARFM model exhibiting the least effectiveness. In September 2022, KFCAM exhibited greater performance—achieving PCC = 0.921, QI = 0.931, R^2 = 0.948, and the lowest RMSE (1.328). In the same month, STGF reached its peak metrics (PCC = 0.894, QI = 0.901, R^2 = 0.927, and RMSE = 1.384), while STARFM recorded its highest R^2 (0.891) and lowest RMSE (1.977); STARFM's PCC (0.862) and QI (0.869) peaked one quarter later, in June 2023 (Fig. 7).

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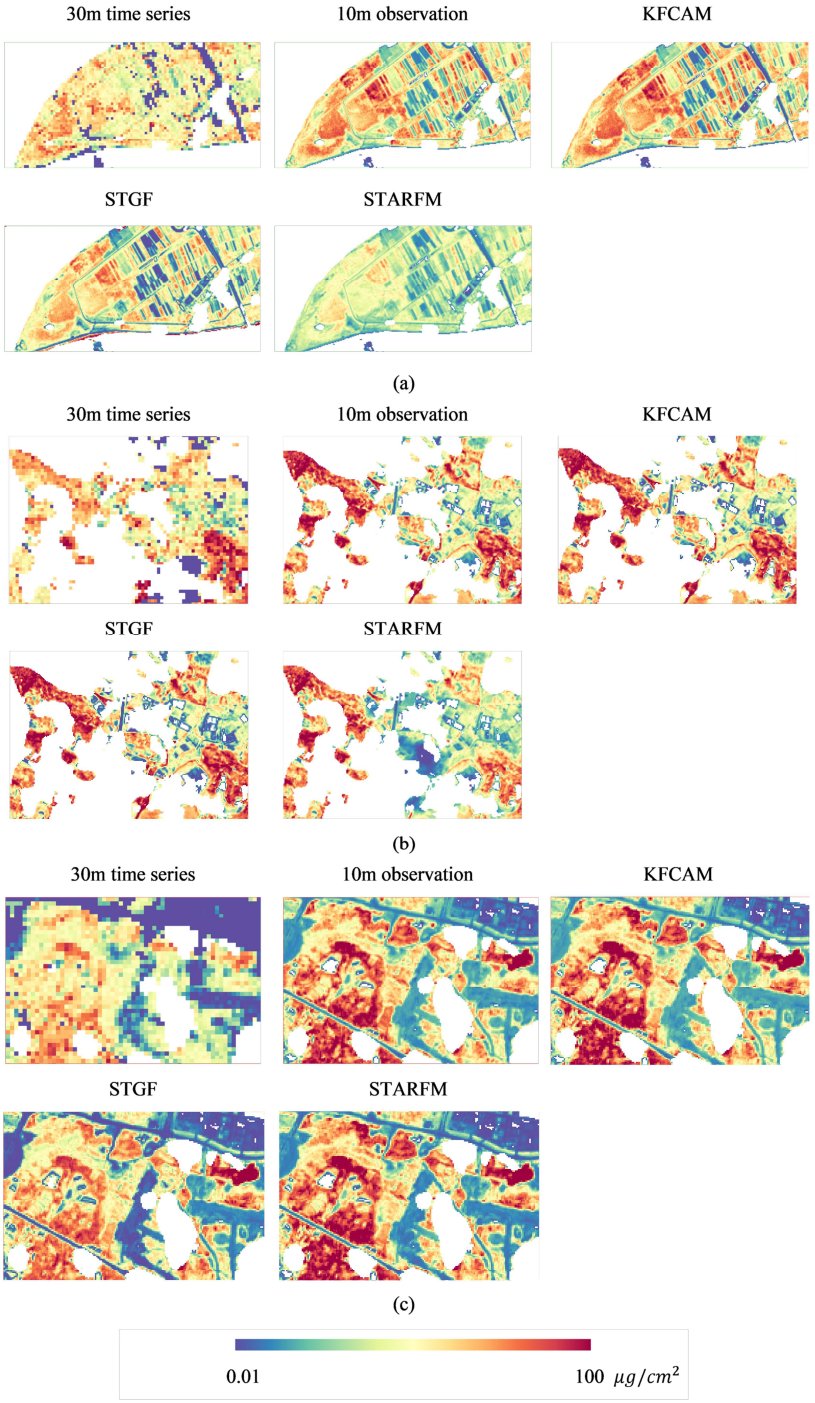


Fig. 5. Comparison between Chl images synthesized by various models and the observed Chl images for September 2022: (a) area1; (b) area2; and (c) area3.

Over the full time series, KFCAM's weakest performance occurred in February 2023 (PCC = 0.871, QI = 0.885, and R^2 = 0.864), with its maximum RMSE (2.179) appearing in April 2023. STARFM registered its lowest R^2 (0.778) in June 2023; furthermore, in April 2023, it produced its minimum PCC (0.779) and QI (0.788), and in February 2023, it produced its highest RMSE (4.332).

Overall, we use the temporal mean as the representative value. The fluctuation range is defined as the maximum absolute deviation from the mean [see (35)]. These metrics are used to evaluate the model's overall performance. KFCAM's RMSE (1.801 $\pm$ 0.369) was approximately 12% lower than that of STGF (2.031 $\pm$ 0.646) and 33% lower than STARFM (2.656 $\pm$ 1.676) (Fig. 7)

$$\begin{aligned} \mathrm{RMSE}_{\mathrm{avg}} &= \overline{\mathrm{RMSE}} \pm D_{\mathrm{RMSE}} \\ D_{\mathrm{RMSE}} &= \left|\max_i \left(\mathrm{RMSE}_i - \overline{\mathrm{RMSE}}\right)\right|. \end{aligned} \tag{35}$$

D. Time-Series Chl Images Assimilated With KFCAM

1) Chl Content Trend: We randomly selected 10 000 points within the experimental area to analyze the Chl content trend.

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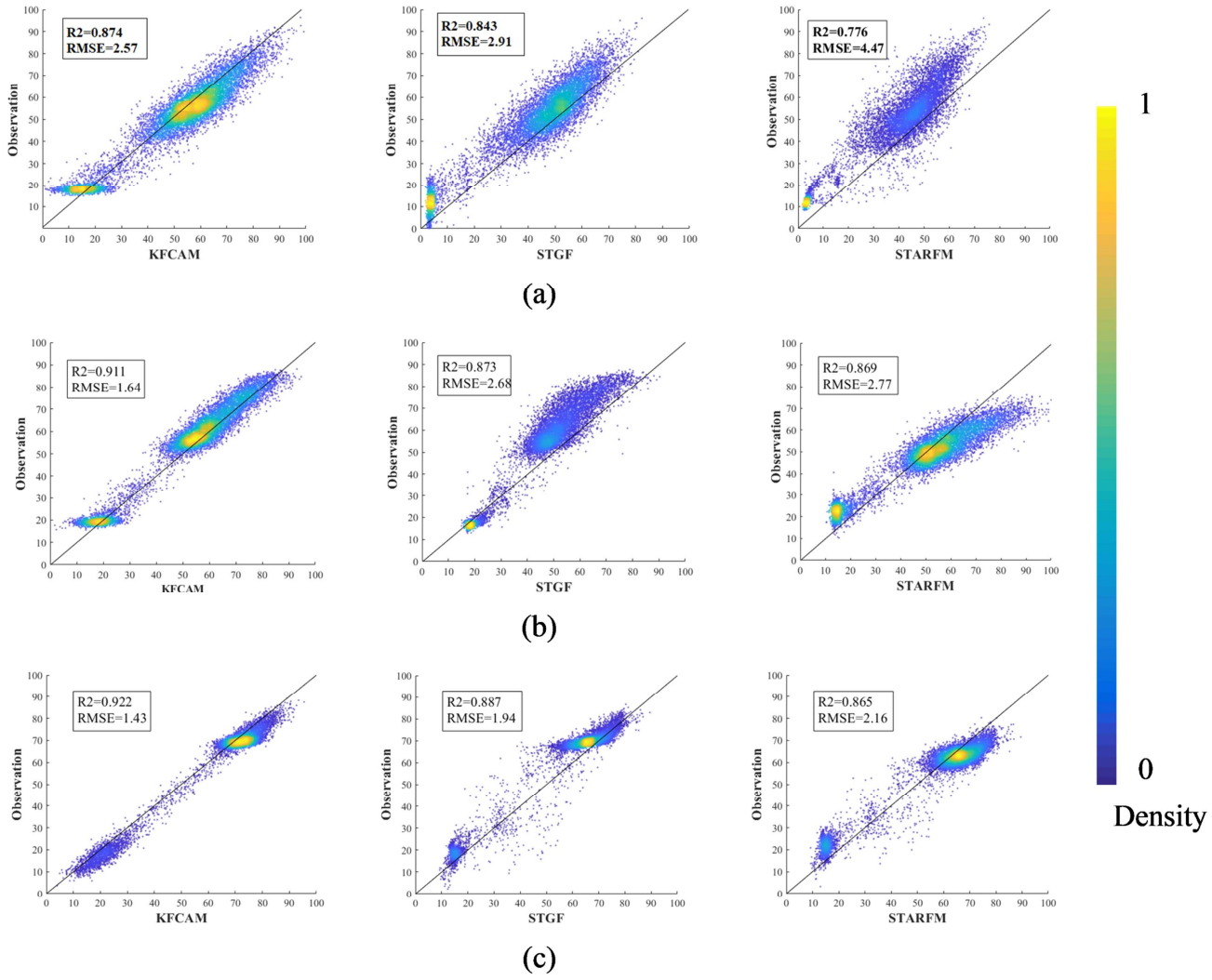


Fig. 6. Scatterplot comparison between model-synthesized Chl images and observed Chl images for September 2022. (a) Results for area 1. (b) Results for area 2. (c) Results for area 3.

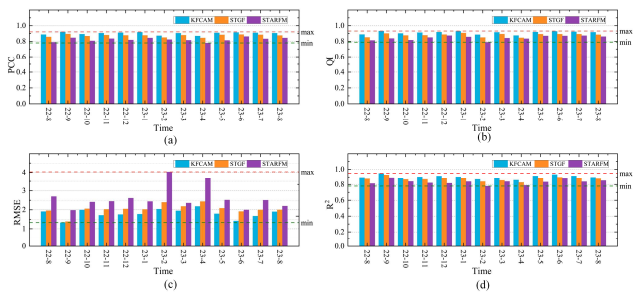


Fig. 7. Comparison of the quality metrics for the KFCAM, STGF, and STARFM models, with (a) representing PCC, (b) representing QI, (c) representing RMSE, and (d) representing R_2.

For each time, the mean value of Chl content was calculated from both model-synthesized Chl images and input images at these selected points, which was then used to plot the time-series curves (Fig. 8).

KFCAM produced a total of 57 Fine-Chl images (10 m), with a time interval of 30 days, which demonstrates an obvious seasonal pattern in temporal variation. Specifically, the Chl contents show an upward trend from February to July each year, while from August to December, it exhibits a downward trend; meanwhile, the study area's annual average Chl value peaks in July and its minimum occurs in January. It is noteworthy that the Chl content distribution assimilated with KFCAM closely aligns with the Fine-Chl images (Sentinel-2/SPOT5), while exhibiting significant bias with the Coarse-Chl images (Landsat-8). These discrepancies were observed during two periods: October 2021–April 2022 and December 2022–February 2023, both across the entire dataset and within the defined small study area. These deviations can be attributed to the substandard data quality of the Landsat-8 data during these specific timeframes. However, in area1 and area2, the results are significantly deviated from the Fine-Chl in July–August 2020 [Fig. 8(b) and (c)]. In contrast, the results in area3 exhibit less deviation overall. This discrepancy may

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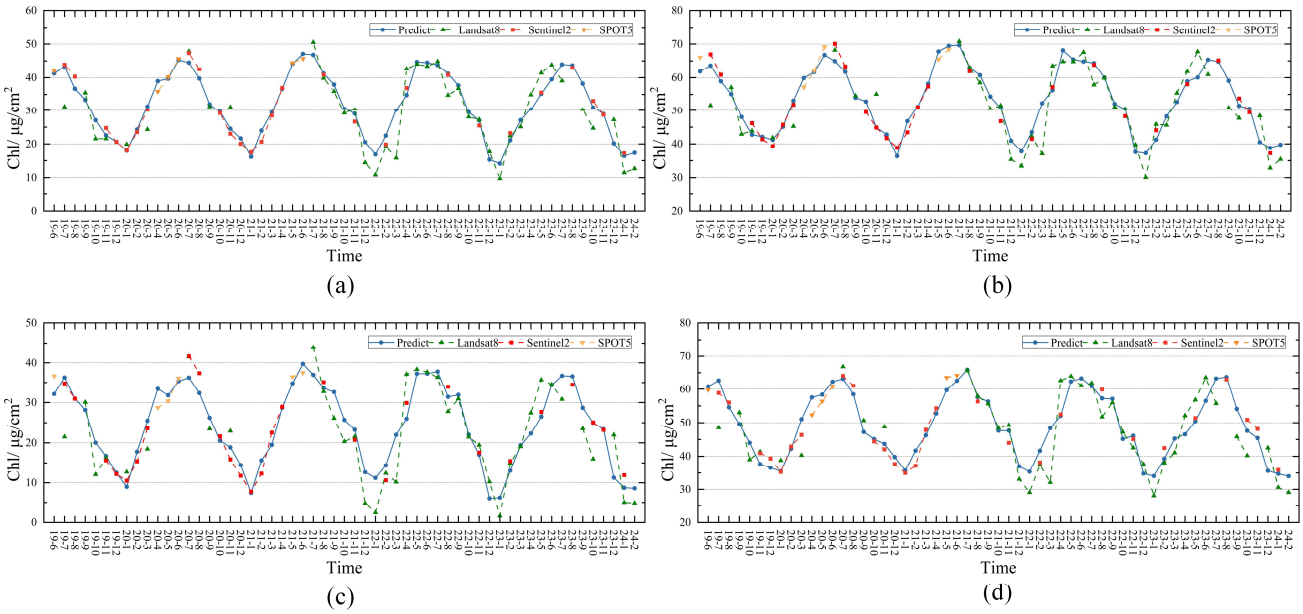


Fig. 8. 30-day Chl time-series curve assimilated with KFCAM in (a) entire study area, (b) area1, (c) area2, and (d) area3.

be related to the coverage of different vegetation types in these areas.

The temporal variation of Chl in grasslands and forests exhibited a similar seasonal variation as the overall study area, with a minimum around January and a maximum around July on an annual basis. Moreover, the mean value of Chl in area3 from September to November 2022 exhibited a substantial decrease, followed by an increase [Fig. 8(d)]. Further investigation is necessary to determine whether this anomaly is attributable to artificial planting. In contrast, the aquatic vegetation predominantly found in area2 displayed a markedly different fluctuation pattern compared to the overall trend. The anomalies primarily occurred during the period of May–July 2020 and August–October 2022 [Fig. 8(c)].

By dynamically adjusting covariance and uncertainty in KFCAM in response to real-time inputs, the predicted Chl state and its associated error are updated when Fine-Chl observations are assimilated. This assimilation produces close agreement between the predicted Chl time series and the observed values from Sentinel and SPOT5 (Fig. 8).

2) Assimilated Chl Content Compared With STGF and STARFM: To provide time-varying rates, it is essential for STGF and STARFM to utilize continuous cross-check time series. Therefore, this experiment focused on the period from August 2022 to August 2023, which features continuous Landsat-8 imagery with one image captured per month. In contrast, Sentinel-2 imagery is available at two-month intervals. To enhance the analysis, monthly clear-sky Sentinel-2 images were used to retrieve Chl images as observational data. The results from KFCAM, STGF, and STARFM were then compared through time-series analysis.

From a time-series perspective, during the initial phase of this study, the Chl synthesized by KFCAM showed a lower degree of similarity to the observed Chl compared to STGF. However, as the assimilation model underwent temporal

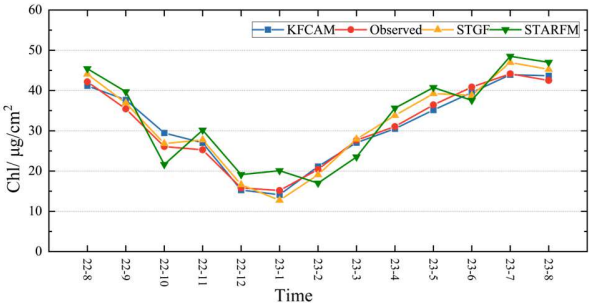


Fig. 9. Temporal comparison of KFCAM, STGF, and STARFM.

optimization, its accuracy improved, resulting in synthesized Chl that aligned more closely with the observed time series. In contrast, the time series generated by STARFM displayed unclear temporal variation patterns and significant fluctuations (Fig. 9). Notably, after March 2023, the KFCAM results exhibited greater alignment with the observed values compared to STGF. The RMSE of the KFCAM model is 1.57, with a maximum of 1.76, while the STGF model has a mean RMSE of 2.07 and a maximum of 2.85. The STARFM model recorded a maximum RMSE of 2.95, the highest among all results (Table IV). In summary, the RMSE of the STGF model is higher than that of the KFCAM and STGF models, indicating that the Chl time series synthesized by KFCAM more precisely captures the true temporal variation characteristics of Chl content. The observed reductions in RMSE and improved temporal stability are attributable not merely to numerical tuning but to the underlying conceptual differences: direct assimilation of Chl (state-space), sensor-aware PROSAIL-LUT observations, and dynamic, data-driven covariance estimation.

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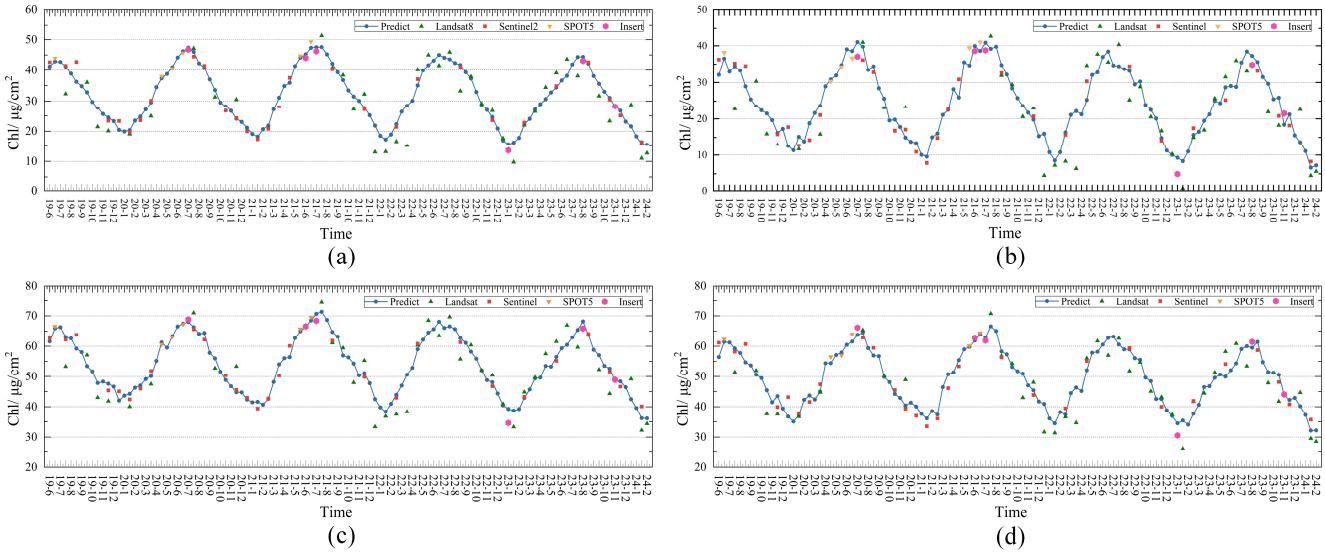


Fig. 10. 15-day Chl time-series curve predicted with KFCAM in (a) entire study area, (b) area1, (c) area2, and (d) area3.

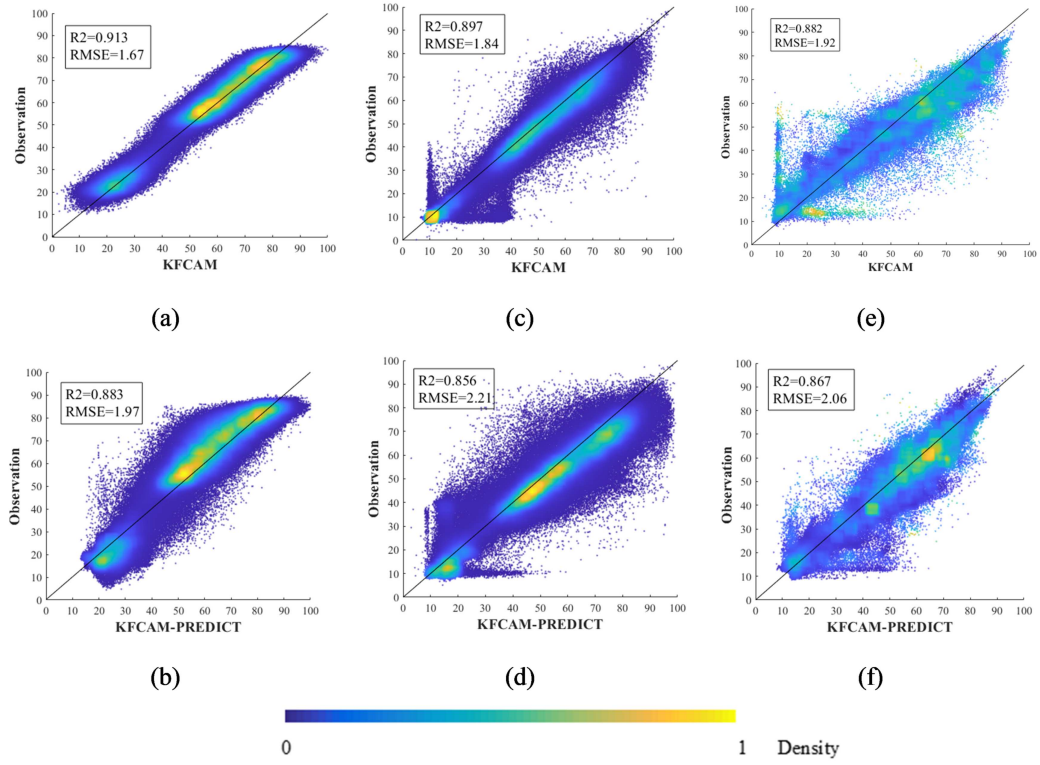


Fig. 11. Comparison of the (a), (c), and (e) KFCAM-synthesized results and (b), (d), and (f) time interpolation prediction results for September 2022, February 2022, and April 2021, respectively.

E. Time-Series Chl Images Predicted With KFCAM

To enhance the temporal resolution of the Chl time series, we reinput 57 previously generated Chl images into KFCAM for new predictions. Using temporal nearest-neighbor interpolation, we obtained 113 Chl images at a 10-m resolution, with one image every 15 days [Fig. 10(a)]. Additionally, six cloud-free Sentinel-2 images were inverted with LUT to update the state estimate.

Although the first assimilation incorporated 35 10-m Fine-Chl images, the second assimilation achieved comparable performance with only six 10-m Fine-Chl images. Furthermore, the new predictions demonstrate the capacity to discern subtle variations in vegetation type [Fig. 10(c) and (d)]. For instance, anomalous fluctuations in aquatic vegetation

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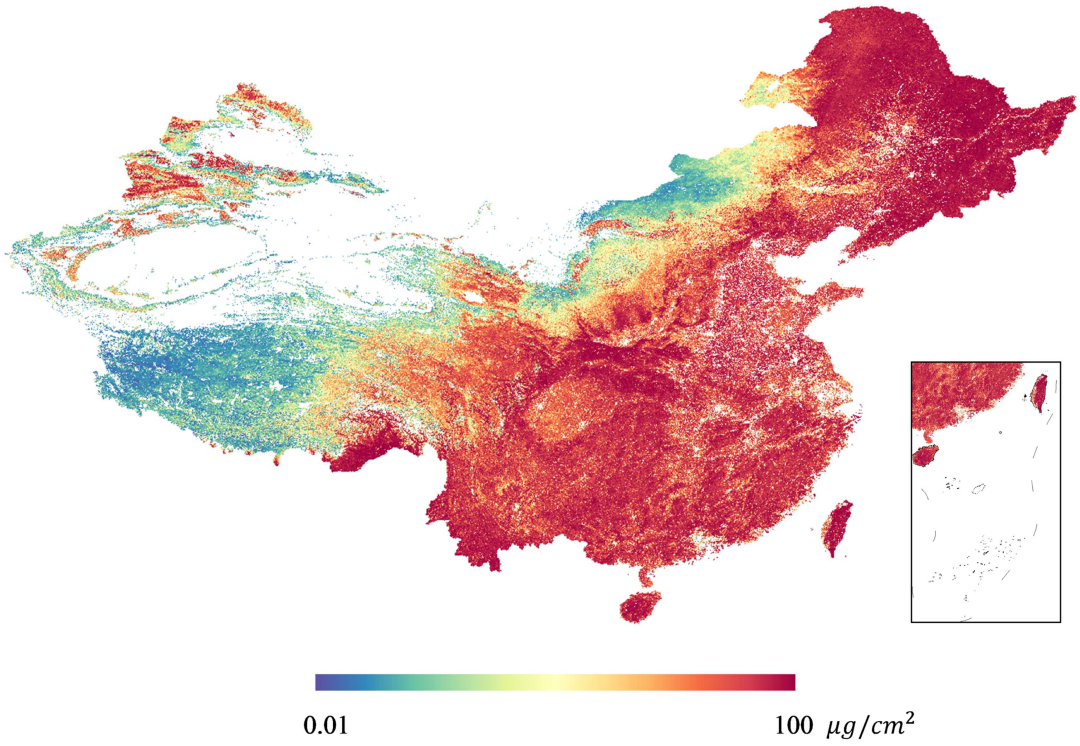


Fig. 12. National-scale Chl content with the well-predicted KFCAM model in 2022.

in area2 during May–July 2020 and June–August 2021 can be clearly identified [Fig. 10(b)]. However, scatterplots comparing the retrieved results with observed data demonstrate that temporal interpolation inevitably leads to some loss of accuracy (Fig. 11). Conversely, at equivalent temporal points, the correlation between the KFCAM-synthesized Chl images and observed data (R^2 = 0.913) exhibits a significantly higher degree of significance in comparison to that of the temporally interpolated results (R^2 = 0.883).

While our experiments focus on Wuhan study area, the KFCAM framework is not location-specific. By design, KFCAM operates on generic state space and observation formulations and therefore can be adapted to other ecosystems and larger spatial extents, provided appropriate sensor harmonization and LUT/RTM parameterization. With systematic cross-sensor calibration and region-appropriate validation, KFCAM can potentially be scaled for regional–global vegetation monitoring to deliver continuous, high-resolution Chl time series.

By applying KFCAM to synthesize Landsat-8, Sentinel-2, and SPOT5 imagery for 2022 across China, we ultimately obtained nationwide Chl content at a resolution of 10 m. In the northwest region of China, remote sensing coverage primarily spans from February to May 2022 (Fig. 12), with some areas in the far northwest covered from May to September. This leads to large regions with low Chl values and smaller areas with high values, which are similar to findings in [57], [58], and [59]. Conversely, the northeastern, central, and southern regions have more comprehensive coverage, primarily from June to August 2022, resulting in higher Chl values. Due to China's diverse geographic characteristics, the distribution of Chl values largely corresponds to the actual situation. However, remote sensing imagery has inherent limitations in providing complete coverage, leading to significant variations that affect the accuracy and detail of the final 10-m Chl image. When multiple images are captured within a time frame, the image with the highest average Chl value is selected as the representative image. The results demonstrate that KFCAM will be applicable to regional and global vegetation monitoring.

IV. Discussion

A. Assessment and Mitigation Strategies for Intersensor Systematic Bias

Differences in sensor SRFs, radiometric resolutions, and acquisition geometries across multisource remote-sensing images can introduce substantial systematic errors. If uncorrected, these sensor-specific discrepancies propagate into the assimilation input and impair model performance. To address this, we applied a PROSAIL-based LUT inversion using a pixel-by-pixel matching strategy prior to assimilation. For every pixel in each image, we matched the pixel's spectral reflectance to the LUT to identify the optimal reflectance–Chl mapping. This pixelwise matching was implemented to minimize intersensor systematic errors in the model input.

We selected two cloud-free high-resolution images of the study area from September 2021 (Sentinel-2 and SPOT5). After strict preprocessing, two Chl images with a spatial resolution of 10 m were obtained via LUT inversion (Fig. 13). Following LUT processing, the two Chl images exhibited very similar spatial distributions. A scatterplot comparison

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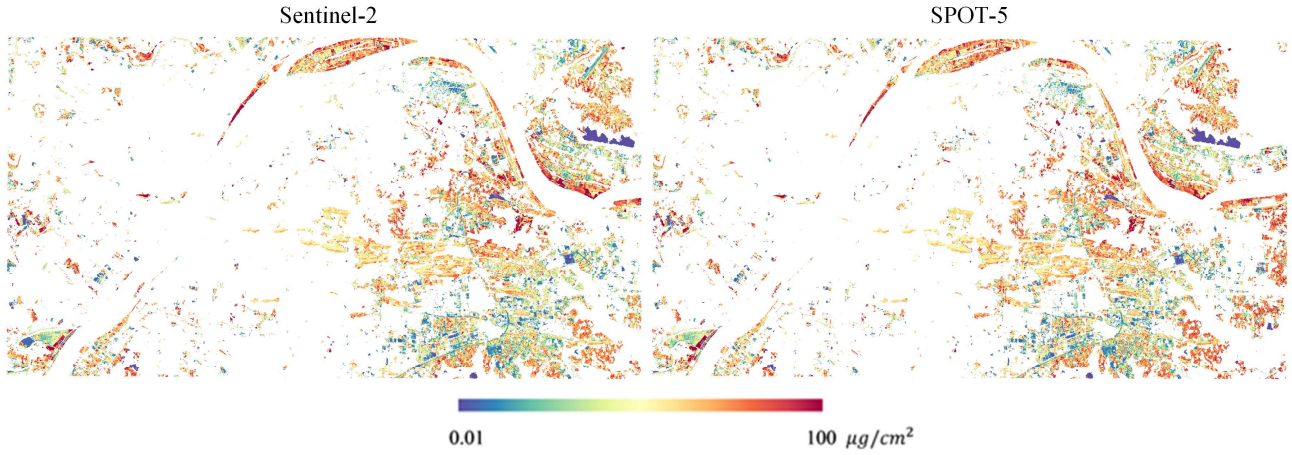


Fig. 13. Chl inversion by LUT of different sensors for September 2021.

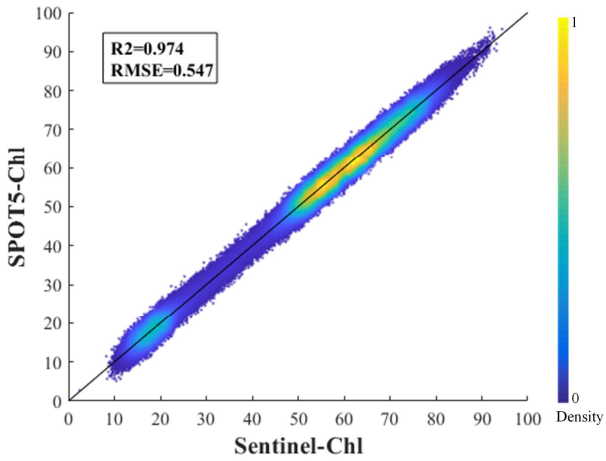


Fig. 14. Comparison of Chl inversion results from different sensor images for September 2021.

showed high agreement (R^2 = 0.974) (Fig. 14), indicating that intersensor systematic error was largely eliminated by the LUT procedure.

After LUT correction, the systematic errors among images were substantially reduced but not entirely eliminated. To assess the influence of these residual biases on the modeled time series and to examine whether validation data selection introduces negative effects on the final results, additional validation experiments were conducted for the period from August 2022 to August 2023. Landsat-8 data were provided monthly, Sentinel-2 images were available in August 2022, February 2023, and August 2023, and SPOT5 images were acquired in November 2022 and May 2023. Two comparative validation experiments were finally performed using monthly clear-sky Sentinel-2 and SPOT5 data to evaluate the impact of systematic bias between the validation datasets and the model observations. Two types of sensor data were used as validation data and synthesized results for error analysis. The difference in RMSE ranged from 0.004 to 0.104 (Table V), indicating minimal deviation. Therefore, for the scales and application scenarios considered in this study, LUT processing reduces systematic bias sufficiently that Kalman data assimilation will not be misled by any remaining bias.

B. Model Sensitivity Across Temporal Intervals

In our assimilation experiment, we initialized four independent runs with a common start date (T_0 = December 2021) and end dates (T_n) in March, May, September, and December 2022—corresponding to assimilation intervals of 3, 5, 9, and 12 months, respectively. Each run incorporated the Sentinel-2 acquisitions from December 2021 and March, May, September, and December 2022, together with all monthly Landsat-8 images during the same period. We, then, generated Chl predictions for January 2023 for subsequent intermodel comparison.

As the assimilation interval lengthened, all three models (KFCAM, STGF, and STARFM) exhibited declining R^2, PCC, and QI alongside increasing RMSE. At a nine-month interval, STARFM and STGF both fell below R^2 = 0.8, whereas KFCAM only dropped below this threshold at 12 months. Notably, extending the interval from three to five months caused KFCAM's RMSE to jump by 0.443. Consequently, we recommend maintaining assimilation windows of three months or less in practical Chl-synthesis applications. Although all models performed worse at the 12-month interval, KFCAM (RMSE = 3.584) still outperformed STGF and STARFM by a substantial margin (see Table VI).

We also observed that, beyond six months, model performance became increasingly insensitive to land-cover variations—especially for STGF and STARFM—underscoring KFCAM's greater temporal stability (Table VI). These results highlight that coarse-to-fine upscaling of Chl (from 30 to 10 m) without regular insertion of high-resolution imagery yields unreliable predictions. To improve long-term forecasting of 10-m Chl fields, it is, therefore, essential to integrate additional high-resolution observations into the KFCAM framework.

C. Model Sensitivity Across Land-Cover Changes

Different types of vegetation cover commonly exhibit subtle differences in how their biochemical parameters change over

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TABLE VI
DIFFERENT TIME INTERVALS OF FINE-CHL IMAGE EFFECTS ON THE MODEL

Model		Time interval			
		3 months	5 months	9 months	12 months
STARFM	R^2	0.863	0.835	0.778	0.701
	PCC	0.853	0.827	0.768	0.707
	QI	0.855	0.845	0.772	0.708
	RMSE	1.669	2.134	3.456	4.197
KFCAM	R^2	**0.894**	0.852	0.811	0.733
	PCC	**0.901**	0.842	0.801	0.739
	QI	**0.901**	0.845	0.818	0.743
	RMSE	**1.514**	1.957	2.353	3.584
STGF	R^2	0.874	0.833	0.791	0.713
	PCC	0.883	0.840	0.782	0.722
	QI	0.880	0.823	0.797	0.707
	RMSE	1.645	2.257	3.015	3.927

TABLE VII
SURFACE COVER CHANGE EXPERIMENT

Model		Time interval			
		1 months	3 months	5 months	7 months
STARFM	R^2	0.745	0.796	0.827	0.836
	PCC	0.752	0.788	0.833	0.830
	QI	0.754	0.791	0.821	0.845
	RMSE	3.804	2.357	2.171	1.894
KFCAM	R^2	0.804	0.823	0.844	**0.851**
	PCC	0.797	0.829	0.837	**0.858**
	QI	0.796	0.833	0.853	**0.858**
	RMSE	2.923	2.103	2.072	**1.779**
STGF	R^2	0.784	0.816	0.832	0.841
	PCC	0.790	0.823	0.823	0.849
	QI	0.789	0.823	0.826	0.832
	RMSE	3.211	2.435	2.204	1.817

the temporal variables [60]. Soil properties and seasonal climatic fluctuations exert primary control over the spatiotemporal dynamics of terrestrial vegetation's biochemical parameters [61], thereby giving rise to pronounced seasonal patterns in Chl concentration. Phytoplankton dynamics and water-quality conditions exert the greatest influence on the biochemical attributes of aquatic vegetation—particularly Chl—during the late-spring to late-summer growth season [62]. These results indicate that, at any given time, both Chl content and its variability differ markedly among vegetation types. Abrupt transitions in land cover during KFCAM assimilation can introduce systematic retrieval errors. To quantify these effects and assess their temporal propagation within the Chl time series, we designed a series of controlled experiments simulating dynamic land-cover scenarios and evaluated the resulting deviations in Chl estimates over successive time steps.

We defined an artificial construction site within the study area that underwent a land-cover transition—from vegetation to bare soil due to construction between May and June 2022—before naturally reverting to grassland. To assess the impact of this abrupt change, we assimilated Fine-Chl images from May, July, September, and November 2022, and January 2023 into the monthly coarse-Chl time series. Predictions were then generated for July, September, November 2022, and January 2023—corresponding to one-, three-, five-, and seven-month intervals following the land-cover change—and compared against the observed Chl images for each date.

The temporal proximity to the land-cover change strongly influenced model accuracy. One month after disturbance, the R^2, PCC, and QI values of STARFM and STGF already fell below 0.8, whereas KFCAM maintained 0.804, demonstrating its greater robustness in rapidly changing environments (Table VII). Although prediction accuracy for all three models improved as the interval from the disturbance increased (evidenced by rising R^2, PCC, and QI and declining RMSE), error induced by the initial land-cover change persisted throughout the time series. Even at a seven-month interval—with Fine-Chl updates every two months—KFCAM's RMSE (1.779) remained higher than its three-month-interval RMSE (1.514; Table VI). Nonetheless, the overall downward trend in RMSE

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TABLE VIII
QUANTITATIVE EVALUATION OF THE EXPERIMENTAL RESULTS USING SENTINEL-2 DATA WITH VARYING CLOUD COVER

Time		Cloud cover					
		1%	5%	10%	20%	30%	50%
May,2023	R^2	**0.882**	0.822	0.733	0.663	0.406	0.244
	RMSE	**1.794**	2.274	5.467	11.457	22.671	39.441
	MCOV	**0.854**	0.801	0.712	0.594	0.401	0.212
Jun,2023	R^2	**0.912**	0.869	0.732	0.696	0.437	0.235
	RMSE	**1.371**	1.985	5.521	9.773	20.934	41.796
	MCOV	**0.901**	0.827	0.717	0.582	0.422	0.205
Jul,2023	R^2	**0.897**	0.837	0.783	0.677	0.458	0.217
	RMSE	**1.879**	2.123	5.211	10.367	20.006	43.147
	MCOV	**0.887**	0.816	0.736	0.581	0.434	0.196
Aug,2023	R^2	**0.905**	0.845	0.763	0.692	0.439	0.287
	RMSE	**1.402**	2.116	5.396	11.074	21.571	35.553
	MCOV	**0.911**	0.813	0.739	0.587	0.421	0.238

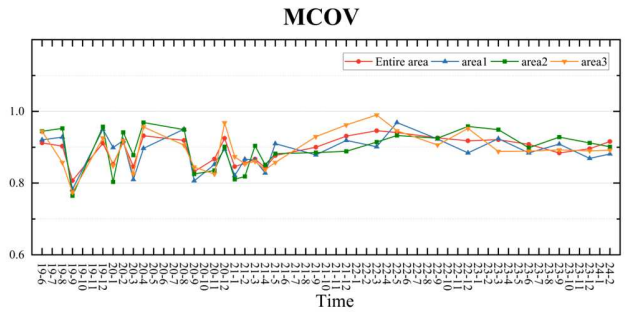


Fig. 15. Prediction uncertainty assessment MCOV metric.

highlights the benefit of repeated high-resolution image insertion and dynamic model updating (Table VII), underscoring KFCAM's capacity to gradually correct disturbance-related errors over time.

D. Uncertainty of KFCAM

KFCAM uses high-resolution Chl images as direct observations, and the observation covariance and process-noise covariance are not fixed hyperparameters but are computed dynamically from the data. Consequently, the model's stability and uncertainty can be quantified at each time step during runtime.

According to (16)–(19), high observation noise in the Kalman filter reduces the Kalman gain, thereby increasing reliance on the model forecast [53]. To quantify this balance—and to evaluate filter stability—we introduce an MCOV metric. At each observation time, during the posterior update (before applying the Kalman gain), we compute the difference between the prior state estimate and the observation and compare it against the observation covariance. MCOV is defined as the fraction of pixels whose innovation falls within this covariance range, yielding values in [1, 0], where values closer to 1 indicate greater weight on the model forecast and, hence, higher filter stability.

In the early stages of prediction, MCOV exhibited irregular fluctuations, dipping to a minimum of 0.8 (Fig. 15), which implies that the Kalman gain dominated the adjustment process and strongly influenced corrections. As assimilation proceeded, MCOV variability diminished and stabilized around 0.9, reflecting increasing operational stability. This pattern held both for the overall study area and for individual subregions—grassland (area 1), aquatic vegetation (area 2), and forest (area 3)—each of which showed large initial swings followed by convergence toward steady, high MCOV values. Subsequent to the stabilization of the model's operation, a corresponding stabilization was observed in the MCOV area, signifying that KFCAM attains robust and consistent performance across diverse vegetation types.

E. Impact of Cloud Cover on KFCAM

Chl is highly responsive to environmental variables, with rapid fluctuations triggered by changes in atmospheric conditions [63]. Consequently, the influence of cloud cover on the quality of synthesized Chl imagery warrants careful consideration. In this study, we observed that the RMSE for September 2022 and June 2023 was markedly lower than in other months [Fig. 7(c)]. A temporal analysis reveals that no Sentinel-2 or SPOT5 scenes were assimilated during these periods; instead, Landsat-8 imagery—characterized by less than 1% cloud cover—dominated the dataset. This finding underscores the strong positive effect of cloud-free observations on fusion accuracy. To quantify the degradation in synthesis performance under increasing cloud cover and to assess the robustness of KFCAM, we conducted a series of comparative experiments. In each assimilation cycle, we incorporated multiple high-resolution Sentinel scenes exhibiting a range of cloud-cover fractions. The outcomes of these optimization experiments are summarized in Table VIII.

The accuracy of synthesized Chl images under different cloud-cover conditions was assessed quantitatively using three performance metrics: R^2, RMSE, and MCOV (Table VII). When cloud cover remained below 10%, the maximum RMSE

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was 5.521 and the minimum R^2 was 0.712, demonstrating that KFCAM can generate Chl images with both high precision and strong explanatory power under low-cloud conditions. However, as cloud cover increased to 20%, the minimum R^2 fell to 0.663, accompanied by substantial information loss in cloud-obscured regions. At cloud-cover levels of 30% and 50%, RMSE rose sharply—reaching 43.147 in the worst case—indicating that data contaminated by high cloud fractions become unsuitable for accurate Chl synthesis. Overall, optimal reconstruction accuracy was achieved when assimilated imagery exhibited less than 10% cloud cover, with a maximum acceptable cloud-cover threshold of 20%.

Fig. 7 further illustrates these dynamics. Under conditions in which Sentinel-2 and SPOT5 images contained under 1% cloud cover and Landsat-8 scenes had roughly 5%, the overall RMSE reached a low of 1.67. By contrast, in the scenarios summarized in Table VI—where all Landsat-8 images were cloud-free while Sentinel-2 data retained 5% cloud cover—the RMSE exceeded 2. This discrepancy highlights that, within the KFCAM framework, the quality of high-resolution imagery used for optimization exerts a greater influence on synthesis accuracy than the quality of low-resolution data employed for constructing the time series.

Table VIII shows that RMSE is highly sensitive to cloud cover: a modest increase in cloud cover from 1% to 5% results in an increase of 0.244–0.714 in RMSE. By contrast, intersensor systematic error ranges only from 0.004 to 0.104. Therefore, cloud cover is the primary error source that must be considered in the assimilation process.

F. Impact of Flowering Season on KFCAM

Flowering season, which occurs annually between January and March [64], represents a critical phenological phase that induces modest peaks in Chl concentration within temporal profiles [8]. However, in Wuhan study area, nursery flower plantings covered only 33 hectares in 2023—approximately 0.04% of the region's land area [65]—limiting their influence on regional Chl dynamics. Among the flowering species cultivated in the study area (Wuhan), cherry blossoms predominate [66]. Except for the atypical green-flowered cultivar Prunus serrulata "Grandiflora," their Chl concentrations remain markedly low [67], and thus, they contribute negligibly to the region's overall Chl distribution. Consequently, seasonal Chl variability in Wuhan is governed primarily by vegetation canopy growth. Peak Chl concentrations align with the maximum canopy development in July, consistent with the phenological patterns reported by Sherman et al. [8].

V. Conclusion

This study introduces a Chl retrieval-assimilation framework based on the KF algorithm (KFCAM), designed to integrate multisource remote sensing data for robust estimation of surface vegetation Chl content. By fusing long-term, low-resolution image time series with high-resolution spatial distribution information, KFCAM effectively addresses common challenges of data gaps and incomplete coverage in remote sensing-based Chl retrieval. We applied KFCAM to generate continuous monthly Chl maps from 2019 to 2024 and evaluated its performance against a reference model in terms of visual fidelity, uncertainty quantification, and temporal consistency. The results demonstrate that: 1) KFCAM achieves high retrieval accuracy ($R^2 > 0.90$); 2) it provides quantitative estimates of model stability and uncertainty during assimilation; 3) it outperforms comparative methods (STGF and STARFM) in scenarios involving long temporal gaps and dynamic land-cover changes; 4) it implemented effective control and quantitative evaluation of model uncertainty and systematic error, demonstrating the method's high accuracy and stability in multisource data assimilation; and 5) it supports subsequent analyses—such as time-series forecasting and spatial interpolation—yielding Chl products with enhanced temporal precision or expanded spatial coverage and it provides a novel means for large-scale, global Chl detection.

Despite these strengths, further refinement of KFCAM is warranted. In particular, defining and quantifying pixel-level uncertainty based on the quality attributes of Landsat and Sentinel input imagery remains an open problem. Additionally, systematic errors and sensor-specific uncertainty associated with fine-resolution platforms (Sentinel-2 and SPOT5) should be rigorously characterized and incorporated into second-generation uncertainty estimates for model outputs. Addressing these topics will be critical for advancing the reliability and applicability of KFCAM in operational vegetation monitoring.

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